Exercises on Matching

From alternating paths to algebraic isolation

How to use this sheet. This is an exercise bank, not a single assignment. The order develops tools before their applications. A $\star$ marks a problem requiring an extra idea; $\star\star$ marks a longer challenge. Sections F–G are optional algebraic and algorithmic extensions. Earlier exercises may be used unless a particular proof method is requested. The accompanying editorial notes record the corrections and the correspondence with the original sheet.

Conventions. Graphs are finite, undirected and simple, except where multigraphs are explicitly allowed. Write $G = (A, B; E)$ for a bipartite graph, with $V(G) = A \dot\cup B$. For a set S contained in one side, $N(S)$ is its set of neighbours on the other side. A matching *saturates* or *covers* a vertex set when every vertex in that set is incident with a matching edge. A *maximum* matching has largest cardinality; a *maximal* matching cannot be enlarged by adding an edge. Write $\nu(G)$ for the maximum matching size, and $\delta(G), \Delta(G)$ for minimum and maximum degree. Path length means the number of edges. The empty graph has a perfect matching, namely the empty matching.

A. Alternating paths and basic structure

1. Maximal versus maximum

Prove that $\nu(G) \le 2|M|$ for every maximal matching M in G. Deduce that $|M_1| \le 2|M_2|$ for any two maximal matchings M_1, M_2. Give a connected graph for which equality holds.

2. Symmetric difference and alternating cycles

Let M, N be matchings in a graph G.

(a) Prove that every component containing an edge of $(V(G), M \triangle N)$ is a path or an even cycle, with edges alternating between M and N.

(b) Deduce that the symmetric difference of two perfect matchings is a vertex-disjoint union of even alternating cycles, possibly empty.

(c) A cycle is *M-alternating* if its edges alternate between edges in M and edges outside M. Show that a perfect matching M is unique if and only if there is no M-alternating cycle.

3. Berge's augmenting-path criterion

An *M-augmenting path* is a nonempty simple path whose two endpoints are not saturated by M and whose edges alternate between edges outside and inside M. Prove that flipping membership in M along such a path increases the matching size by one. Then prove that a matching is maximum if and only if it has no augmenting path. This exercise is not restricted to bipartite graphs.

4. Combining two saturation requirements

Let $G = (A, B; E)$, $S \subseteq A$ and $T \subseteq B$. Suppose one matching saturates S and another saturates T.

(a) Use symmetric difference to construct a matching saturating $S \cup T$.

(b) Show that such a matching can be chosen to be maximum in G.

5. The edge version of Slither

Two players alternately choose edges of a nonempty graph. The first player chooses any edge. Thereafter a move must extend the current simple path at *either* endpoint by one edge to a vertex not yet on the path. The first player unable to move loses. Prove that the first player has a winning strategy whenever the graph has a perfect matching. Describe an invariant for the path after each move of the first player. The edge-starting convention is part of the question; see [And74].

B. Hall's theorem, deficiency and duality

6. Hall's theorem from alternating paths

Let $G = (A, B; E)$. Starting with a matching M that has no augmenting path, orient unmatched edges from A to B and matching edges from B to A.

(a) Suppose some vertex of A is unmatched. Let $S \subseteq A$ and $T \subseteq B$ be the vertices reachable from unmatched vertices of A. Prove that $N(S) = T$ and $|S| > |T|$.

(b) Deduce *Hall's theorem*: G has a matching saturating A if and only if $|N(X)| \geq |X|$ for every $X \subseteq A$.

7. Systems of distinct representatives

Let $F_1, \ldots, F_r$ be finite sets. A *system of distinct representatives* is a choice of pairwise distinct elements $x_i \in F_i$. Translate this problem into bipartite matching and prove that such a system exists if and only if

$$\left|\bigcup_{i \in I} F_i\right| \geq |I| \qquad \text{for every } I \subseteq \{1, \ldots, r\}.$$

Explain why the condition includes the cases of an empty set F_i and $I = \varnothing$ without any additional convention.

8. Kőnig's theorem and a second proof of Hall

A *vertex cover* meets every edge. Write $\tau(G)$ for its minimum size.

(a) Let M be a maximum matching in a bipartite graph. Using the reachable sets S, T from Exercise 6, including the case that there are no unmatched vertices in A, show that $C = (A \setminus S) \cup T$ is a vertex cover with $|C| = |M|$. Deduce Kőnig's theorem $\nu(G) = \tau(G)$.

(b) Starting only from Kőnig's theorem and elementary counting, derive Hall's theorem. Do not invoke the conclusion of Exercise 6 in this second proof.

9. The deficiency form of Hall's theorem

For $X \subseteq A$, define

$$\operatorname{def}(X) = |X| - |N(X)|, \qquad d = \max_{X \subseteq A} \operatorname{def}(X).$$

Since $\operatorname{def}(\varnothing) = 0$, we have $d \geq 0$. Show that every matching leaves at least d vertices of A unmatched. For the reverse bound, add d new vertices to B, each adjacent to all of A, and apply Hall's theorem. Conclude that

$$\boxed{\nu(G) = |A| - d.}$$

10. Submodularity, supermodularity and uncrossing

For all $X, Y \subseteq A$, prove

$$|N(X \cup Y)| + |N(X \cap Y)| \leq |N(X)| + |N(Y)|,$$

and hence

$$\mathrm{def}(X \cup Y) + \mathrm{def}(X \cap Y) \geq \mathrm{def}(X) + \mathrm{def}(Y).$$

Thus the neighbourhood-size function is *submodular*, whereas def is *supermodular*. Show that the sets attaining maximum deficiency are closed under union and intersection. When Hall's condition holds, call a set X *tight* if $|N(X)| = |X|$. Show that tight sets are closed under union and intersection, and that for two tight sets

$$N(X \cap Y) = N(X) \cap N(Y).$$

C. Degrees, density and applications of Hall's theorem

11. Regular bipartite graphs and decomposition

Let $G = (A, B; E)$ be a nonempty r-regular bipartite graph, where $r \geq 1$. Use edge counting and Hall's theorem to prove that $|A| = |B|$ and that G has a perfect matching. Deduce that its edge set is the disjoint union of r perfect matchings. Extend both conclusions to bipartite *multigraphs*, where parallel edges are counted with multiplicity.

12. Choosing cards from columns

There are mn cards, one for each pair consisting of one of m values and one of n colours, where $m, n \geq 1$. Arrange the cards arbitrarily in an array with n rows and m columns.

(a) Prove that one can choose one card from every column so that the m chosen cards have distinct values.

(b) Strengthen this: partition all the cards into n groups, each containing one card from every column and one card of every value. A bipartite multigraph is useful here. No restriction on the colours within a chosen group is imposed.

13. Completing a Latin rectangle

An $r \times n$ *Latin rectangle* has entries in $\{1, \ldots, n\}$, with no repeated entry in a row or a column. Let $0 \leq r < n$. Prove that it can be extended by one row to an $(r+1) \times n$ Latin rectangle, and hence completed to an $n \times n$ Latin square. Identify the relevant regular bipartite graph and compute its degree.

14. Saturating all vertices of maximum degree

Let $G = (A, B; E)$ have at least one edge, put $\Delta = \Delta(G)$, and let $W = \{v : \deg(v) = \Delta\}$.

(a) Use Hall's theorem to find a matching saturating $A \cap W$.

(b) Apply the analogous statement on B, together with Exercise 4, to find one matching saturating all of W.

(c) Is the analogous assertion for the vertices of *minimum* degree true? Give a counterexample with no isolated vertices.

15. Bipartite edge colouring

A proper edge colouring gives different colours to edges with a common endpoint. Prove that every bipartite graph has a proper edge colouring using $\Delta(G)$ colours, with the edgeless case interpreted separately. Use Exercise 14 to remove a matching that reduces the maximum degree, and argue by induction. Why can fewer colours not suffice when there is at least one edge?

16. A local degree comparison ⋆

Let $G = (A, B; E)$ have no isolated vertex in A. Suppose that

$$\deg(a) \geq \deg(b) \qquad \text{for every edge } ab, \quad a \in A,\ b \in B.$$

Prove that G has a matching saturating A. Connectivity is not assumed. One possible approach is to give each edge ab a contribution $1/\deg(a)$ and count the total contribution from a set $S \subseteq A$ at its neighbours.

17. An extremal edge bound

Let $G = (A, B; E)$ with $|A| = |B| = n \geq 1$. Prove that $|E| \leq n\nu(G)$. Deduce that, for every integer $1 \leq k \leq n$,

$$|E| > (k-1)n \quad \Longrightarrow \quad \nu(G) \geq k.$$

Show that the strict inequality cannot be weakened: construct a graph with $(k-1)n$ edges and no matching of size k.

18. A minimum-degree condition for a perfect matching

Suppose $G = (A, B; E)$, $|A| = |B| = n \geq 1$, and $\delta(G) \geq n/2$. Prove that G has a perfect matching by checking Hall's condition for small and large subsets of A separately. For each $n \geq 2$, give a graph with minimum degree $\lceil n/2 \rceil - 1$ and no perfect matching.

19. Checking Hall only on small sets on both sides ⋆

Let $|A| = |B| = n \geq 1$, and let $p, q \geq 0$ be integers with $p + q = n$. Assume $|N(S)| \geq |S|$ whenever $S \subseteq A$, $|S| \leq p$, and also whenever $S \subseteq B$, $|S| \leq q$.

(a) Prove that G has a perfect matching. Start from a hypothetical Hall obstruction in A and consider vertices in B outside its neighbourhood.

(b) Show that the threshold is sharp: for every $p, q \geq 0$ with $p + q = n - 1$, construct a balanced bipartite graph satisfying the two stated small-set conditions but having no perfect matching.

20. Doubly stochastic matrices ⋆

Let $X = (x_{ij})$ be an $n \times n$ real matrix, $n \geq 1$, with nonnegative entries and every row sum and column sum equal to 1.

(a) Use Hall's theorem to find a permutation π such that $x_{i,\pi(i)} > 0$ for every i.

(b) Prove that X is a convex combination of permutation matrices: $X = \sum_{j=1}^{t} \lambda_j P_j$, where $\lambda_j > 0$ and $\sum_j \lambda_j = 1$. Give a finite procedure by repeatedly subtracting a suitable multiple of a permutation matrix. Justify termination, including the case when the remaining matrix is itself a permutation matrix.

D. Allowed edges, uniqueness and trees

21. Strict Hall inequalities and extendable edges

Let $G = (A, B; E)$ satisfy

$$|N(S)| \geq |S| + 1 \qquad \text{for every } \varnothing \neq S \subsetneq A.$$

(a) Prove that every edge belongs to a matching saturating A. Fix an edge ab and apply Hall's theorem after deleting a and b. Do not assume that $|A| = |B|$.

(b) Suppose additionally that G is connected, $|A| = |B| \geq 1$, and every edge belongs to a perfect matching. Prove the strict inequalities above. Explain the role of connectivity.

22. An exact Hall obstruction to using an edge

Assume that $G = (A, B; E)$ has a matching saturating A. Call an edge *allowed* if it belongs to some matching saturating A. Prove that $ab \in E$, $a \in A$, is *not* allowed if and only if there is a tight set $S \subseteq A \setminus \{a\}$ with $b \in N(S)$. In particular, show that every A-saturating matching matches a tight set S bijectively onto $N(S)$.

23. A vertex with no forbidden incident edge ⋆

Let $A \neq \varnothing$, and suppose $G = (A, B; E)$ has a matching saturating A. Prove that there exists $a \in A$ such that *each* edge incident with a belongs to *some* maximum matching of G. The maximum matching may depend on the edge. Use Exercise 22 and an inclusion-minimal nonempty tight set, treating the case of no such tight set separately. Can the conclusion be strengthened to hold for every vertex of A?

24. Directed reachability and unique perfect matchings ⋆

Let $M = \{a_i b_i : 1 \leq i \leq n\}$ be a perfect matching of a bipartite graph, where $n \geq 1$. Form a directed graph D on $\{1, \ldots, n\}$, putting an arc $i \to j$ for each edge $a_i b_j \notin M$.

(a) Show that $a_i b_j \notin M$ belongs to a perfect matching if and only if there is a directed path from j to i. Equivalently, i, j lie in the same strongly connected component of D.

(b) Prove that M is unique if and only if D is acyclic. Deduce that in this case the rows and columns of the biadjacency matrix can be ordered so that it is upper triangular with diagonal entries 1.

(c) Show that a bipartite graph with a unique perfect matching has a vertex of degree 1 on each side. Deduce a leaf-removal procedure for recovering its perfect matching.

25. Perfect matchings in trees ⋆

For a graph H, let $\operatorname{odd}(H)$ be the number of its connected components having an odd number of vertices. Let T be a tree with at least one vertex.

(a) Show that T has at most one perfect matching.

(b) Prove the characterization

$$T \text{ has a perfect matching} \quad \Longleftrightarrow \quad \operatorname{odd}(T - v) = 1 \text{ for every } v \in V(T).$$

Give a proof that does not invoke Tutte's theorem.

(c) Under the condition on the right, let each vertex v point to its neighbour in the unique odd component of $T - v$. Show that the resulting choices pair vertices with one another and recover the perfect matching.

26. A greedy algorithm for maximum matching in a forest

Let v be a leaf of a forest F and let u be its neighbour. Show that some maximum matching contains uv, and deduce

$$\nu(F) = 1 + \nu(F - \{u, v\}).$$

Design a linear-time algorithm that repeatedly processes leaves and isolated vertices. Explain how to use it to decide whether F has a perfect matching. Specify how degrees and the queue of available leaves are maintained.

E. Reductions, approximation and augmenting-path algorithms

27. Reducing prescribed coverage to perfect matching

This exercise concerns arbitrary graphs, not only bipartite graphs.

(a) Given $G = (V, E)$ and $S \subseteq V$, construct a graph H such that H has a perfect matching if and only if G has a matching saturating $V \setminus S$. Use at most $2|V| + 1$ vertices, prove both directions, and explicitly handle parity and the cases $S = \varnothing, V$.

(b) Given $0 \le k \le \lfloor |V|/2 \rfloor$, reduce the question $\nu(G) \ge k$ to one perfect-matching decision problem. Try adding $|V| - 2k$ independent new vertices, each adjacent to every original vertex.

28. Search from decision: what can be done in parallel?

Suppose an oracle decides whether a given graph has a perfect matching.

(a) Give a procedure using at most $|E(G)| + 1$ oracle calls that finds a perfect matching or reports that none exists. One option is to delete an edge whenever deletion preserves the existence of a perfect matching.

(b) Why is it incorrect to test all edge deletions on the original graph and then delete every edge whose individual deletion is safe? Give a four-vertex counterexample.

(c) Under the promise that the perfect matching is unique, show that one batch of edge-deletion queries does identify all its edges. Explain why the procedure in part (a) is adaptive whereas this one is not.

29. Short augmenting paths and approximation

Let M be a matching in an arbitrary graph and let $k \ge 1$ be an integer. Suppose there is no M-augmenting path of length at most $2k - 1$. Prove that

$$|M| \ge \frac{k}{k+1}\nu(G), \qquad \nu(G) - |M| \le \frac{|V(G)|}{2k+2}.$$

Compare M with a maximum matching using symmetric difference. For each k, exhibit a path showing that the multiplicative factor is best possible. Recover Exercise 1 when $k = 1$.

30. The Hopcroft–Karp phase argument ⋆⋆

In a bipartite graph with $N \geq 1$ vertices and m edges, a *phase* augments along an inclusion-maximal collection of vertex-disjoint shortest augmenting paths.

(a) Prove that a phase strictly increases the shortest augmenting-path length, unless the matching becomes maximum. Use breadth-first-search layers.

(b) Prove that $O(\sqrt{N})$ phases suffice. Once shortest augmenting paths are long, bound the deficit using symmetric difference with a maximum matching.

(c) Implement a phase in $O(m + N)$ time using breadth-first and depth-first search. Deduce the $O((m + N)\sqrt{N})$ time bound [HK73].

F. Matrices, determinants and matching polynomials

31. The symbolic biadjacency matrix

Let $G = (A, B; E)$. Assign a distinct indeterminate x_e to each edge and form the $|A| \times |B|$ matrix $B_G(\mathbf{x})$ whose ab entry is x_{ab} when $ab \in E$ and 0 otherwise. Work over an arbitrary field $\mathbb{F}$.

(a) When $|A| = |B|$, show that $\det B_G(\mathbf{x})$ is a nonzero polynomial if and only if G has a perfect matching. Explain why distinct perfect matchings cannot cancel, even in characteristic 2.

(b) Prove that $\operatorname{rank}_{\mathbb{F}(\mathbf{x})} B_G(\mathbf{x}) = \nu(G)$ by examining minors.

(c) Explain why assigning $x_e = 1$ to every edge need not preserve the rank. Give an example with two vertices on each side.

32. All square minors of a forest adjacency matrix

Let F be a forest and let A_F be its ordinary 0–1 adjacency matrix, viewed first as an integer matrix. Let $R, C \subseteq V(F)$ with $|R| = |C|$. Form a bipartite graph $H_{R,C}$ on a row-copy of R and a disjoint column-copy of C, with an edge $r_{\mathrm{row}}c_{\mathrm{col}}$ exactly when $rc \in E(F)$.

(a) Prove that $H_{R,C}$ is a forest. It is a subgraph of a graph consisting of two disjoint copies of F.

(b) Prove that $\det A_F[R, C] \in \{0, 1, -1\}$, and that it is nonzero exactly when $H_{R,C}$ has a perfect matching. Thus A_F is *totally unimodular*.

(c) For a principal submatrix, show that

$$\det A_F[R, R] = \begin{cases} (-1)^{|R|/2}, & F[R] \text{ has a perfect matching,} \\ 0, & \text{otherwise.} \end{cases}$$

(d) Deduce that $\operatorname{rank}_{\mathbb{F}} A_F = 2\nu(F)$ over every field $\mathbb{F}$.

33. The matching polynomial of a forest ⋆

For a forest F on N vertices, let $m_k(F)$ be the number of its k-edge matchings, with $m_0(F) = 1$. Use the determinant expansion to prove

$$\chi_F(t) := \det(tI - A_F) = \sum_{k=0}^{\lfloor N/2 \rfloor} (-1)^k m_k(F) t^{N-2k}.$$

Derive the recurrence, for every vertex v,

$$\chi_F(t) = t\chi_{F-v}(t) - \sum_{u \in N(v)} \chi_{F-\{u,v\}}(t), \qquad \chi_{\varnothing}(t) = 1.$$

Over $\mathbb{R}$, show that the multiplicity of the root 0 is $N - 2\nu(F)$. Give a counterexample to recovering $\nu(F)$ this way after reduction modulo a prime. Explain why this does not contradict the rank identity in Exercise 32.

34. Randomized testing through determinants

You may use the polynomial zero bound: a nonzero polynomial of total degree at most d, evaluated at independent uniform elements of a finite set $Q \subseteq \mathbb{F}$, vanishes with probability at most $d/|Q|$.

(a) For a balanced graph with n vertices on each side, independently replace each x_e in Exercise 31 by a uniform element of Q, where $|Q| > n$. Prove that testing the resulting determinant gives a one-sided-error test for perfect matching, with failure probability at most $n/|Q|$. State precisely which answer can be erroneous.

(b) For a rectangular symbolic biadjacency matrix, prove that the sampled rank never exceeds $\nu(G)$ and equals $\nu(G)$ with probability at least $1 - \nu(G)/|Q|$. It is enough to fix one nonzero minor of maximum size; a union bound over all minors is unnecessary.

G. Weighted matchings and isolation

For this section, the weight of a matching is $w(M) = \sum_{e \in M} w(e)$. A weight assignment is *isolating* when there is a unique minimum-weight perfect matching. Whenever a minimum-weight perfect matching is discussed, its existence is explicitly assumed.

35. Nonzero circulation forces uniqueness

For an even cycle $C = (e_1, \ldots, e_{2r})$ in cyclic order, define

$$\mathrm{circ}_w(C) = \left| \sum_{i=1}^{2r} (-1)^{i-1} w(e_i) \right|.$$

(a) Show that this definition is independent of the starting point and direction of traversal.

(b) Let a bipartite graph G have a perfect matching and let $w : E \to \mathbb{R}$. Prove that if every cycle has nonzero circulation, then the minimum-weight perfect matching is unique. Compare two optimal matchings *one alternating cycle at a time*.

(c) Generalize the argument to an arbitrary graph by requiring nonzero circulation only for every *nice even cycle*: an even cycle C for which $G - V(C)$ has a perfect matching [DKR10].

36. The union of all minimum-weight perfect matchings **

Let $G = (A, B; E)$, with $|A| = |B| \geq 1$, have a perfect matching. Let $w : E \to \mathbb{R}$, and let μ be the minimum perfect-matching weight. Let G_w have vertex set $V(G)$ and exactly the edges occurring in at least one minimum-weight perfect matching.

(a) Choose finitely many optimal matchings $M_1, \ldots, M_r$ whose union is G_w, and retain repeated edges as parallel copies. Explain why their multiset union is an r-regular bipartite multigraph.

(b) Prove that *every* perfect matching of G_w has weight μ. Use Hall's theorem through the decomposition in Exercise 11.

(c) Prove that *every* cycle C in G_w satisfies $\mathrm{circ}_w(C) = 0$, even when C is not alternating with respect to any individual perfect matching. Work with the multigraph in part (a), adjusting multiplicities on alternate edges of C. No linear-programming duality is needed; see [FGT16, Section 3.1].

(d) Explain why bipartiteness matters. Consider two disjoint triangles $a_1a_2a_3a_1$ and $b_1b_2b_3b_1$, joined by edges a_ib_i. Give the joining edges weight 1 and the triangle edges weight 0. Determine its optimal-support graph, and find a perfect matching and an even cycle that refute the two corresponding general-graph claims.

37. Zero circulation and vertex potentials ⋆

Let $H = (A, B; E)$ be a connected bipartite graph with real edge weights. Prove that all cycles have zero circulation if and only if there are functions $\alpha : A \to \mathbb{R}$ and $\beta : B \to \mathbb{R}$ such that

$$w(ab) = \alpha(a) + \beta(b) \qquad \text{for every } ab \in E.$$

Use a spanning tree for the nontrivial direction. Deduce that all perfect matchings of H have the same weight. Apply the conclusion componentwise to the graph G_w from Exercise 36.

38. The Isolation Lemma ⋆

Let $\mathcal{F}$ be a nonempty family of subsets of $\{1, \ldots, m\}$, where $m \geq 1$. Independently assign to each element a uniform weight in $\{1, \ldots, L\}$, where $L \geq 1$. Prove that

$$\Pr[\mathcal{F} \text{ has a unique minimum-weight member}] \geq 1 - \frac{m}{L}.$$

Condition on all weights except one, and compare the best set containing that element with the best set avoiding it. Carefully handle a comparison class that is empty. For a graph with at least one edge and a perfect matching, deduce the $1/2$ success bound for isolation by taking $L = 2|E|$ [MVV87].

39. Recovering an isolated matching from determinants ⋆⋆

Let $G = (A, B; E)$, $|A| = |B| = n \geq 1$, have a unique minimum-weight perfect matching M^* under weights $w : E \to \mathbb{Z}_{\geq 0}$. Define $B_w(t)$ to have entry $t^{w(ab)}$ on an edge and 0 elsewhere, and put $P(t) = \det B_w(t)$. For a nonzero polynomial, ord_t denotes its smallest occurring exponent; set $\operatorname{ord}_t 0 = +\infty$.

(a) Prove that $\operatorname{ord}_t P = w(M^*)$, and that the corresponding coefficient is 1 or -1. This statement is valid over every field.

(b) Let $P_e(t)$ be the analogous determinant after deleting edge e. Prove that

$$e \in M^* \quad \Longleftrightarrow \quad \operatorname{ord}_t P_e > \operatorname{ord}_t P.$$

Explain why all these tests can be made independently.

(c) Now evaluate at $t = 2$ over the *integers*. With $v_2(0) = +\infty$, show that $v_2(P(2)) = w(M^*)$ and that the same edge-recovery rule works with v_2 in place of ord_t. State why integer arithmetic, rather than arithmetic in characteristic 2, is required in this part [MVV87].

40. Large isolating weights and the cost of their encoding ⋆

Enumerate the edges as $e_0, \ldots, e_{m-1}$ and set $w(e_i) = 2^i$.

(a) Prove that distinct edge subsets have distinct total weights. Deduce isolation whenever a perfect matching exists, and also prove that every even cycle has nonzero circulation.

(b) Each weight has only $O(m)$ bits. Nevertheless, explain why the polynomial matrix in Exercise 39, represented by dense coefficient lists, and the integer matrix obtained by setting $t = 2$ can require exponentially much space in m.

(c) Explain why this construction alone does not yield a polynomial-size determinant computation in either of those two representations. Distinguish the bit length of $w(e)$, the degree of $t^{w(e)}$, and the bit length of $2^{w(e)}$.

Sources and provenance

The starting point was the supplied two-page *Exercises on Matching* (20 numbered questions). All of its topics are retained, with corrections, reordering and additional scaffolding. The separate editorial notes give the full old-to-new map and identify the 18 entirely added exercises; "added" refers to this sheet, not to originality of the underlying result. The references below locate the named game and the advanced algorithmic results.

References

[And74] William N. Anderson, Jr. *Maximum Matching and the Game of Slither*. Journal of Combinatorial Theory, Series B **17** (1974), 234–239. doi:10.1016/0095-8956(74)90029-X.

[HK73] John E. Hopcroft and Richard M. Karp. *An $n^{5/2}$ Algorithm for Maximum Matchings in Bipartite Graphs*. SIAM Journal on Computing **2**(4) (1973), 225–231. doi:10.1137/0202019.

[MVV87] Ketan Mulmuley, Umesh V. Vazirani and Vijay V. Vazirani. *Matching is as easy as matrix inversion*. Combinatorica **7** (1987), 105–113. doi:10.1007/BF02579206.

[DKR10] Samir Datta, Raghav Kulkarni and Sambuddha Roy. *Deterministically Isolating a Perfect Matching in Bipartite Planar Graphs*. Theory of Computing Systems **47** (2010), 737–757. arXiv:0802.2850.

[FGT16] Stephen A. Fenner, Rohit Gurjar and Thomas Thierauf. *Bipartite Perfect Matching is in quasi-NC*. Proceedings of STOC 2016, 754–763. arXiv:1601.06319; see Lemma 2.2, Lemma 3.2 and Corollary 3.3.